

Material Traceability Report: 316L Stainless Steel Flat Washers - Selective Laser Melting Process

Report ID: TR-316L-WSH-2403-18

Date: March 18, 2024

Batch Reference: SLM-BATCH-316L-033

Product: 316L Stainless Steel Flat Washers (33 units)

Manufacturing Process: Selective Laser Melting (SLM)

Quality Assurance Manager: Elena Rodriguez

Executive Summary

This material traceability report documents the complete material flow for production lot SLM-BATCH-316L-033, comprising 33 units of 316L stainless steel flat washers manufactured via selective laser melting. The production cycle demonstrates a material utilization efficiency of 14.8% for finished parts, with significant material recovery through powder reuse and water recycling systems. All materials have been tracked from initial receipt through processing to final disposition, maintaining full chain of custody documentation.

1.0 Material Receiving and Verification

1.1 Raw Material Incoming Inspection

The production cycle commenced with receipt of virgin stainless steel material for powder feedstock production. All materials underwent incoming quality verification per QSP-08-114.

Raw Material Receipt Record:

Material Description	Supplier	Certificate No.	Received Quantity	QA Status
X2CrNiMo1712 Stainless Steel (316L base)	MetalAlloy Inc.	CERT-MA-88732	4.11 kg	Approved
Process Water (Atomization Grade)	Plant Supply	N/A	16.8 kg	Approved

The X2CrNiMo1712 material certificate confirms compliance with ASTM A276/A276M-17 specifications for 316L stainless steel composition, with chemical analysis verifying maximum 0.03% carbon, 16-18% chromium, 10-14% nickel, and 2-3% molybdenum content.

2.0 Powder Feedstock Production

2.1 Water Atomization Process

The virgin stainless steel material was processed through the water atomization system to produce the powder feedstock required for SLM manufacturing. The atomization process parameters were maintained within established operating windows.

Water Atomization Material Balance:

Material Flow	Quantity	Disposition
Input - Stainless Steel	4.11 kg	To atomization
Input - Process Water	16.8 kg	Cooling media
Output - 316L Powder	3.09 kg	To SLM process
Output - Recovered Water	16.4 kg	To recycling system
Output - Solid Waste	0.41 kg	To landfill

Note: Mass balance shows 0.01 kg variance within acceptable 0.25% tolerance for process accounting.

2.2 Energy Consumption for Atomization

The water atomization process consumed 2.55 kWh of electrical energy for the complete powder production cycle. This energy consumption aligns with historical performance data for similar batch sizes.

Historical Comparison: Previous quarter average for equivalent powder production: 2.58 kWh (within 1.2% variance).

3.0 Selective Laser Melting Manufacturing

3.1 Process Setup and Parameters

The SLM manufacturing cycle was configured for production of 33 flat washer components in a single build cycle. Process parameters were established per established manufacturing procedures.

SLM Process Configuration:

Parameter	Value	Unit
Working Cycle Duration	13.38	hours
Components per Cycle	33	pieces
Nominal Power (Laser Active)	5.5	kW
Nominal Power (Laser Inactive)	3.5	kW
Argon Volume per Component	54	liters
Chamber Fill Volume	700	liters

3.2 Material Consumption and Energy Usage

The SLM process consumed powder feedstock and shielding gas while converting electrical energy to manufacture the final components.

SLM Process Inputs:

Material/Energy Type	Quantity	Purpose
316L Powder Feedstock	3.09 kg	Component fabrication
Argon Shielding Gas	3.08 kg	Atmosphere control
Electrical Energy	64.92 kWh	Process power

The argon consumption was calculated based on established process parameters including chamber purging and continuous shielding during the 13.38-hour processing cycle.

3.3 Finished Product Output

The manufacturing cycle successfully produced the complete batch of flat washer components meeting all dimensional and material specifications.

Product Output Summary:

Product Description	Quantity	Total Mass	Quality Status
316L Flat Washers	33 units	0.61 kg	100% Conforming

All components passed dimensional inspection per drawing WSH-316L-02 rev. C and visual examination per acceptance standard VIS-08.

4.0 Material Recovery and Waste Management

4.1 Powder Recovery and Reuse

Following the SLM build cycle, unused powder was systematically recovered from the build chamber and processing equipment. The powder underwent sieving and quality verification before disposition.

Powder Recovery Disposition:

Powder Stream	Quantity	Disposition
Reused in SLM Process	2.94 kg	Returned to powder handling system
Returned for Remelting	0.15 kg	To water atomization feedstock
Non-recyclable Waste	0.01 kg	To landfill

The high powder reuse rate (95.1% of recovered powder) demonstrates effective powder management practices and maintains material within the manufacturing loop.

4.2 Water Recovery and Waste Streams

The water atomization process incorporated closed-loop water recovery, significantly reducing freshwater consumption and waste generation.

Water and Waste Management:

Stream	Quantity	Management Method
Recovered Process Water	16.4 kg	Closed-loop recycling
Solid Waste (Atomization)	0.41 kg	Landfill disposal
Powder Waste (SLM)	0.01 kg	Landfill disposal

The water recovery rate of 97.6% reflects efficient process water management and supports environmental compliance objectives.

5.0 Complete Mass Balance

The comprehensive material accounting for the complete manufacturing cycle demonstrates traceability from raw material input through to finished products, recovered materials, and waste streams.

Complete Material Balance Summary:

Material Category	Input (kg)	Output (kg)
Raw Materials		
Stainless Steel	4.11	-
Process Water	16.8	-
Argon Gas	3.08	-
Products		
Finished Washers	-	0.61
Recovered Materials		
Reused Powder	-	2.94
Remelt Powder	-	0.15
Recovered Water	-	16.4
Waste Streams		
Solid Waste	-	0.41
Powder Waste	-	0.01
TOTALS	23.99	20.52

Note: The mass balance shows expected variance due to gas consumption and process losses, with overall material accountability within acceptable limits for manufacturing traceability.

6.0 Traceability and Compliance

6.1 Chain of Custody Documentation

All material movements are documented through the following reference documents: - Material Receipt Records: MRR-240318-01 through MRR-240318-03 - Process Work Orders: WO-SLM-240318-15 - Quality Inspection Reports: QIR-316L-240318-22 - Waste Manifests: WM-240318-07, WM-240318-08

6.2 Material Utilization Metrics

Key Performance Indicators: - Finished Part Yield: $0.61 \text{ kg} / 4.11 \text{ kg} = 14.8\%$ - Total Material Recovery: $(2.94 + 0.15 + 16.4) \text{ kg} / 23.99 \text{ kg} = 81.2\%$ - Powder Utilization Efficiency: 3.09 kg powder consumed for 0.61 kg finished product

These metrics align with established benchmarks for SLM manufacturing of small precision components and demonstrate continuous improvement over previous reporting periods.

7.0 Conclusions and Recommendations

The material traceability analysis confirms complete documentation of all material flows for production lot SLM-BATCH-316L-033. The manufacturing process achieved target output with appropriate material recovery and minimal waste generation. All materials are fully accounted for with maintained chain of custody from receipt through final disposition.

Recommendations: 1. Continue current powder management practices to maintain high reuse rates 2. Monitor water recovery system performance for potential optimization 3. Document argon consumption trends for future process improvements

This report provides comprehensive material traceability supporting quality assurance requirements and regulatory compliance obligations.

Prepared by: Michael Chen, Production Engineer

Reviewed by: Elena Rodriguez, Quality Assurance Manager

Approved by: David Thompson, Operations Director

Distribution: Quality Records, Manufacturing File, Regulatory Compliance